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Spectroscopic evidence of irreversible changes in cesium loaded iron phosphate glasses under pressure

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ABSTRACT

Detailed Raman spectroscopic studies on 40:60 iron phosphate glasses (IPG) and cesium doped IPG under ambient conditions revealed an evident disproportionation of the pyrophosphate network beyond 18 mol% Cs doping. In situ high pressure Raman spectroscopic studies were carried out on IPG and Cs doped IPG up to 40 GPa for the first time in order to investigate the structural stability of these glasses under pressure. Although the parent glass is found to be robust to such high pressures, pressure quenched samples loaded with cesium beyond 18 mol% concentration were observed to permanently cross link leading to new high density forms of the glass. The extent of cross linking is observed to systematically increase with the cesium content in the parent matrix.

1. Introduction

Iron phosphate glasses — 40 mol% Fe_2O_3:60 mol% P_2O_5 (IPG) are currently being explored as alternative matrices for waste streams that are poorly suited to be vitrified in borosilicate glass matrices [1–6]. Their excellent chemical durability combined with low processing temperatures makes them very good candidates for immobilizing volatile radioactive wastes. High level nuclear wastes (HLW) containing volatile radioactive cesium are known to generate large amounts of thermal energy that would limit the concentration of these wastes that could be immobilized in glassy matrices [7]. Iron phosphate glasses containing 40 mol% Fe_2O_3 are found to be very effective in vitrifying such HLW rich in radioactive cesium sludges [7]. Predicting the long term behaviour of these glasses intended for high-level long-lived nuclear waste requires a thorough understanding of glass structure alteration effects under simulated waste loading conditions. Optic methods like Raman and infrared have always proved to be useful tools in following the structural modifications of these glasses associated with waste loading [3,7–13]. In the present work we have carried out detailed Raman measurements on IPG and IPG loaded with varying amounts of Cs_2O as a simulated waste and studied the associated changes in the structure as Cs is being loaded in the matrix. Studying the structural stability of these Cs loaded glasses under hostile conditions is essential for a comprehensive assessment of the viability of these glasses in nuclear waste disposal [14–18]. In view of this, we present here the structural stability of IPG and Cs loaded IPG to very high external pressures. Although extensive studies have been carried out on the effect of pressure on silica glasses, very few reports exist on the pressure effects of phosphate glasses [16,19–21]. Most of the above studies are restricted to investigations of mechanical properties of these glasses that would serve to enhance wear resistance. Since the dynamics under high pressure conditions can be studied by following the related vibrational processes, optical methods like Raman and infrared spectroscopy prove to be valuable tools to elucidate the associated structural changes [14–16]. In this context we had earlier investigated the structural stability of IPG under a peak pressure of 25 GPa using Raman spectroscopy [16]. In the present work we have extended the Raman spectroscopic data of IPG to still higher pressures up to ~40 GPa and have obtained some of the first high pressure Raman data for IPG and cesium loaded IPG.

2. Experimental details

Iron phosphate glasses of nominal composition $40Fe_2O_3 \cdot 60P_2O_5$ and cesium loaded IPG were prepared by melting appropriate quantities of $NH_4H_2PO_4$ (99.5% Merck, source of P), Fe_2O_3 (99.99% Alfa Aesar, source of Fe) and Cs_2CO_3 (99.9% Aldrich, source of Cs) at their required pouring temperatures in an alumina crucible. The melts were then swirled several times and air quenched onto stainless steel dies. Care was taken to ensure a Fe/P ratio of 0.67 and an O/P ratio of 3.5 in all the samples in order to maintain the pyrophosphate stoichiometry. Table 1 gives the sample identification code, the nominal composition of each of the constituents in the glass sample in mol% along with the respective pouring temperatures. Estimated error on nominal composition values reported in Table 1 is ~3% [22].

Powder XRD measurements were carried out on IPG and cesium loaded IPG using an APD 2000 PRO X-ray diffractometer (GNR Analytical Instruments) using Cu Kα radiation in the Bragg Brentano geometry. X-ray diffraction measurements confirmed the amorphous nature of the

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Table 1
Nominal batch compositions in mol% along with pouring temperatures. Estimated error on nominal composition values is ~3% [22].

Sample code	Nominal glass composition (mol%)			Pouring temperature (K)
	Fe_2O_3	P_2O_5	Cs_2O	
IPG	40	60	0	1423
IPGC5	38	57	5	1403
IPGC12	35	53	12	1373
IPGC18	33	49	18	1353
IPGC26	30	44	26	1323
IPGC28	28	43	28	1283
IPGC35	26	39	35	1263

undoped and all the cesium doped samples (Fig. 1). No signature of devitrification has been observed even for IPGC35. This is in agreement with earlier studies on cesium doped IPG [22]. Raman spectra under ambient conditions were measured using a high throughput micro-Raman spectrometer (Renishaw — model inVia) in the backscattering geometry using 25 mW power of the 514 nm line of an Ar ion laser. A charge coupled device (CCD) was used for detecting the scattered light. In situ high pressure Raman measurements (ambient to ~40 GPa with no pressurizing medium) were carried out using a compact symmetric diamond anvil cell with anvils of a 500 μm culet diameter. IPG glass chips of lateral dimensions of ~100 μm were loaded along with tiny crystals of ruby that served as a pressure calibrant into a 200 μm diameter hole drilled in a pre-indented stainless steel gasket. Pressure was measured by the calibrated shift of the R1 fluorescence line of ruby. Samples were allowed to soak at the respective pressures for 10 min prior to recording the spectra. For all in situ measurements, care was taken to ensure that the Raman spectroscopic data was obtained under the same laser power, focusing conditions and collection time. All Raman spectra were background subtracted, area normalized and fitted to a mixture of Lorentzian and Gaussian line shapes. Good counting statistics ensured that the fitting χ^2 values were close to 1. Fitting errors from analysis are shown for all derived quantities in respective figures.

3. Results

3.1. Raman spectra of IPG and Cs doped IPG under ambient conditions

Fig. 2 shows the set of area normalized Raman spectra of IPG and cesium loaded IPG under ambient conditions. This set of spectra agrees very well with earlier reports [7,11]. Raman spectrum of undoped IPG reveals all the expected phonon modes and is in good agreement with earlier works (Fig. 2) [9–13,16,18,23–26]. The spectral features of cesium doped glasses are easily understood in terms of systematic deviations from the spectrum of the undoped base glass with increasing cesium content. Detailed peak fit analysis of the modes was carried out and the extracted line shape parameters were plotted as a function of cesium concentration to understand the changes associated with the glass network as cesium enters the matrix. Fig. 3 represents the variation of frequency and normalized area of fitted components with cesium loading. Fig. 4 depicts the extent of disproportionation of the IPG network on account of cesium doping.

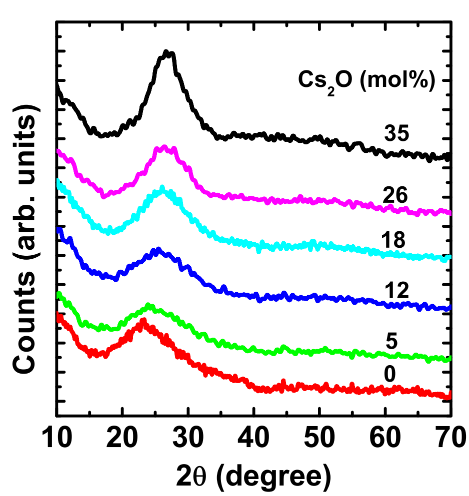

Fig. 1. XRD pattern of IPG and cesium loaded IPG. No signature of divitrification is observed even for IPGC35.

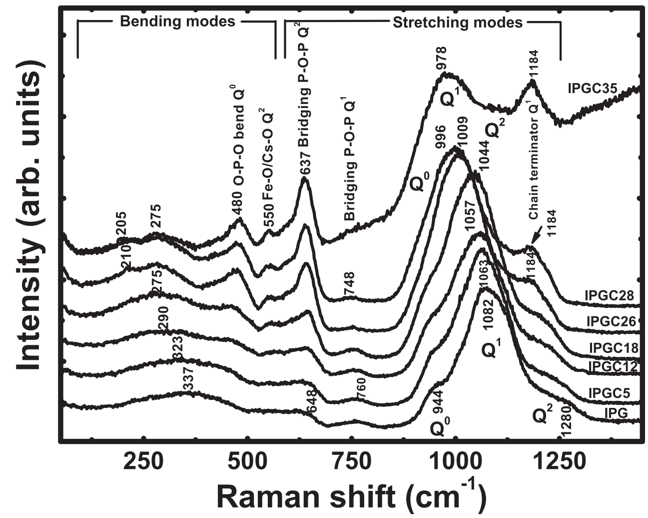

Fig. 2. Set of area normalized Raman spectra of IPG and all cesium loaded analogues under ambient conditions. Note a systematic down-shift of the Q^1 stretch mode at 1082 cm^{-1} in IPG with increased cesium loading that points to a progressive weakening of the glass network.

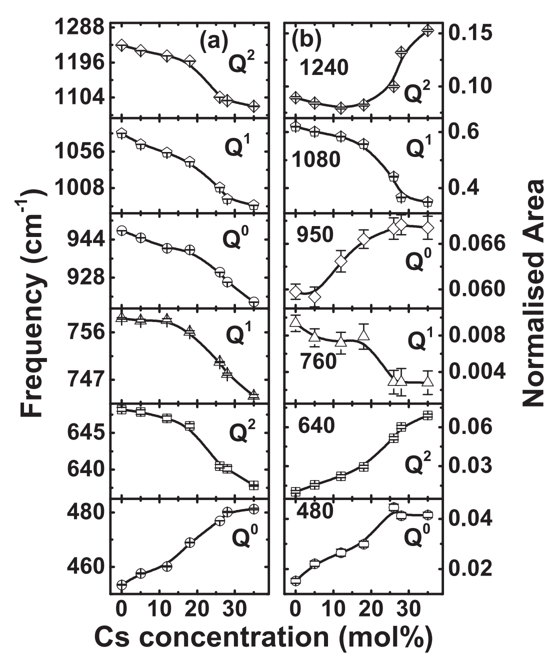

Fig. 3. (a) Frequency variation of the modes corresponding to Q^0, Q^1 and Q^2 units with increased cesium loading. All the stretching modes except the 480 cm^{-1} bending mode are observed to soften, pointing to a progressive weakening of the network on cesium loading. Note an evident change in slope in the frequency trends for all the samples beyond 18 mol% cesium loading. (b) Variation of normalized area of the constituent units with cesium content. Suppression of area corresponding to Q^1 units along with an increase in area of Q^0 and Q^2 units beyond 18 mol% cesium concentration is clearly evident. Lines are guide to the eye.

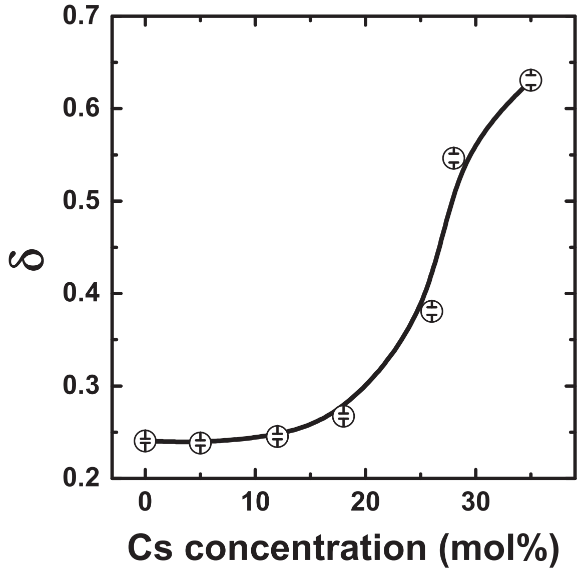

Fig. 4. Variation of the disproportionation factor $\delta = [\mathrm{Area}(Q^0) + \mathrm{Area}(Q^2)] / \mathrm{Area}(Q^1)$ with cesium content. Solid line is guide to the eye. Note the sudden increase beyond 18 mol% cesium concentration clearly pointing to the disintegration of the parent glass network beyond this composition.

3.2. High pressure studies on IPG and cesium loaded IPG

Fig. 5a shows the set of Raman spectra of undoped IPG recorded in-situ as a function of increasing pressure (Ambient to 41 GPa) at room temperature. The plotted spectra reveal pressure induced changes both in the spectral position of the bands and the Raman intensities. For pressure greater than 20 GPa, all the modes appear totally smeared and featureless. It is indeed very interesting to note from the figure that on quenching the pressure from ~41 GPa to ambient pressure the bands regain their intensities and revert back to their original positions. This elastic response of IPG is in sharp contrast to the behaviour of other tetrahedral glasses where permanent densification and structural transformations occur at much lower pressures [27–31]. A similar ambient temperature, in-situ pressure run was carried out on IPGC18 spanning a pressure range between ambient and 39 GPa (Fig. 5b). On decompression, all the modes are found to once again revert back to their original positions whilst minor changes in intensities are observed in the pressure quenched spectra when compared to the ambient.

High pressure Raman studies were carried out on heavily doped IPGC26, IPGC28 and IPGC35 by exposing them to peak pressures of 41, 39 and 32 GPa respectively. Fig. 6 shows the evolution of the Raman spectra of IPGC26 and IPGC35 with pressure. The high pressure Raman data of IPGC28 is not shown in the figure. Quenching these samples from their respective peak pressure results in a significantly altered spectrum in each case, both with respect to spectral intensity and position, clearly revealing an irreversible pressure induced transformation. In the following sections, we try to understand the local structural changes of IPG upon Cs loading and as exposed to high external pressures.

4. Discussions

4.1. Raman spectra of IPG

The most intense Raman band at 1082 cm^{-1} in undoped IPG, as shown in Fig. 2, represents the [P–O] stretching frequency of the Q^1 tetrahedra (containing a single bridging oxygen) of the $(P_2O_7)^{4-}$ pyrophosphate units that form the major building units in these glasses. The shoulders at both the low- and high-frequency ends of this main feature, peaked at 944 and 1280 cm^{-1}, correspond to the stretching vibrations of the isolated Q^0 tetrahedra of the $(PO_4)^{3-}$ units and the Q^2 tetrahedra of the $(PO_3)^-$ units, respectively. The broad and low intensity band at 760 cm^{-1} is assigned to the symmetric stretch of the P–O–P bridges connecting the phosphate tetrahedra in the pyrophosphate structures. The 648 cm^{-1} feature is assigned to the bridging P–O–P stretch of Q^2 units. The low-frequency bands between 300 and 600 cm^{-1} represents the O–P–O bending vibrations of the $(PO_4)^{-1}$ network. The band at 337 cm^{-1} is assigned to the bending vibrations of the Q^0 units with iron as the modifier forming Fe–O polyhedra [23].

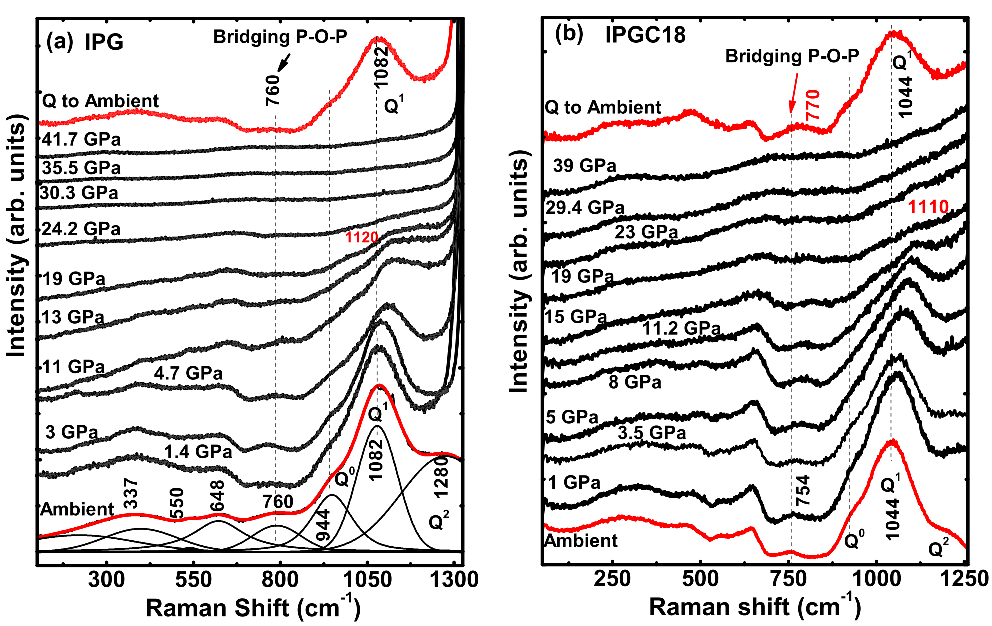

Fig. 5. (a) In-situ high pressure Raman spectra of IPG (Ambient—41 GPa) along with the decompressed spectrum. Resolved peaks are shown for ambient spectrum of IPG sample. Note the excellent reversion of all the modes on decompression. (b) Set of in-situ high pressure Raman spectra of IPGC18 (Ambient—39 GPa) along with decompressed spectrum. All the modes in the decompressed spectrum agree well with the ambient modes except for marginal changes in intensity. Vertical lines are guide to eye.

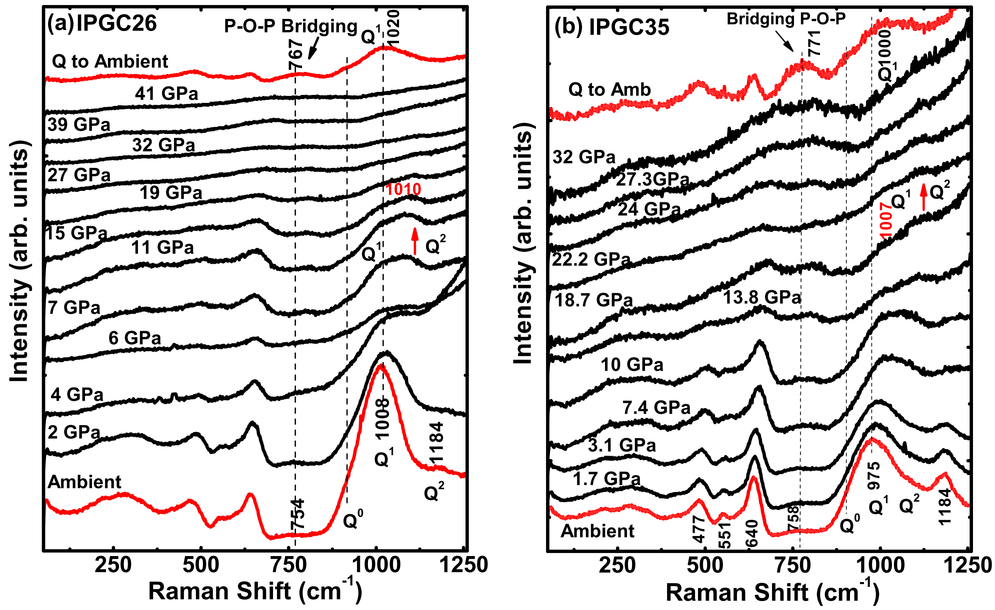

Fig. 6. (a) In-situ high pressure Raman spectra of IPGC26 from ambient to 41 GPa along with their decompressed spectrum. Note the anomalous increase in Q^2 units for pressures greater than 6 GPa. Decompression from the highest peak pressure results in a significantly altered spectrum. A significant increase in spectral weight is observed for the 760 cm^{-1} mode in pressure quenched sample. (b) In-situ high pressure Raman spectra of IPGC35 from ambient to 32 GPa along with their decompressed spectrum. An evident increase in the Q^2 intensity is seen at pressures greater than 6 GPa. Vertical lines are guide to eye. The increased area under the 760 cm^{-1} band implies extensive cross linking in the pressure quenched sample as compared to the ambient spectrum. See text.

4.2. Raman spectra of Cs doped IPG

Correlated changes of the Raman phonons were observed as cesium was progressively loaded into the matrix. The low frequency bending modes <450 cm^{-1} show an evident softening as cesium enters the matrix (Fig. 2). The bending mode at 337 cm^{-1} in the base glass is seen to systematically downshift to 275 cm^{-1} in IPGC18, after which one can note a splitting of this mode into a doublet. The softening of this mode could mean an increase in the O – (P/Fe) – O polyhedral bond angle arising due to a weakening of the network with the incoming cesium. The splitting of this mode could possibly indicate a change in the site symmetry occurring on account of tetrahedral distortions associated with the incoming cesium. New modes at 480 cm^{-1} and 550 cm^{-1} corresponding to the bending modes of Q^0 and Q^2 units respectively are observed to emerge and gradually gain in intensity with increased cesium doping. The bridging P – O – P stretch modes of the Q^2 units at 648 cm^{-1} develop into a well defined feature, whilst the corresponding mode at 760 cm^{-1} for the Q^1 units is found to gradually decrease and finally collapse for the maximum cesium loaded sample. The dominant spectral feature at 1082 cm^{-1} arising due to the symmetric stretch of the pyrophosphate units (Q^1 units) that make up the majority of the network in this glass along with the low and high frequency shoulders corresponding to Q^0 and Q^2 units is found to systematically down-shift to lower energies with increased cesium doping. This clearly indicates a progressive weakening of the entire glass network caused by the scission of the bridging oxygen links leading to depolymerisation of the pyrophosphate dominated network [7,10,11]. A new mode at 1184 cm^{-1} is seen to emerge in the 26 mol% Cs doped sample (IPGC26) due to terminal P – O stretch of Q^1 chain terminator [32]. This mode is observed to progressively increase in intensity for higher levels of cesium loading. It is known that for initial doping levels the incoming cesium aids in cross linking [11] whilst beyond 18 mol% cesium, the alkali metal oxide can effectively break down the extended polymeric network. This terminates the dimeric Q^1 species, resulting in the production of terminal P – O units. In keeping with the above explanation, this mode is observed to weakly emerge in IPGC26 and then systematically increase in intensity for samples with increasing cesium content. It is to be mentioned that this feature becomes a prominent mode in IPGC35, wherein a maximum disruption of the pyrophosphate network is expected.

In order to understand the changes in the base glass matrix associated with cesium loading in a detailed manner, deconvolution of the broad spectral features into the constituent components were carried out. The modes were fitted to a mixture of Gaussian and Lorentzian line shapes and the trends in frequency and area of the constituent modes were plotted as a function of cesium content (Fig. 3a and b). Fig. 3a shows the frequency variation of the modes higher than 450 cm^{-1} as a function of increased cesium loading. All the high frequency stretch modes are observed to soften whilst the 480 cm^{-1} bend mode is observed to harden on increased cesium doping. The 480 cm^{-1} mode is sensitive to changes in the intra-tetrahedral bond angle of the isolated Q^0 units. The hardening of this mode can be understood as follows. It is known that doping the system with cesium leads to depolymerisation of the network. This results in the production of isolated Q^0 units wherein the phosphate tetrahedra are expected to be more compact with lower intra tetrahedral bond angles compared to the Q^1 or Q^2 units leading to the observed hardening of this mode. The softening of the rest of the stretch modes arises due to the overall weakening of the PO_4 tetrahedral network on doping with cesium. A significant change in slope in the frequency trends of all the modes is noted beyond 18 mol% cesium doping that could be attributed to a disproportionation of the glass network. It is known that the presence of high field strength modifying cations in the matrix tends to disproportionate the pyrophosphate linkages that make up the majority of the units of the base glass into meta Q^2 and ortho Q^0 units beyond this composition ($2Q^1 \rightarrow Q^2 + Q^0$) [7,11,26,32,33]. Additional evidence pointing to the disproportionation of the pyrophosphate network for doping beyond IPGC18 can be got from a plot of the variation of normalized area of the modes with the cesium content (Fig. 3b). A clear picture of depletion of the area corresponding to the Q^1 units along with an evident increase in the area of Q^2 and Q^0 units beyond 18 mol% cesium doping is observed. Further the extent of disproportionation, defined by the disproportionation factor $\delta = [\text{Area}(Q^0) + \text{Area}(Q^2)] / \text{Area}(Q^1)$, is seen to more or less remain a constant up to 18 mol%. Beyond this concentration a sudden upturn in the curve is observed, implying extensive disproportionation of the glass network (Fig. 4). Hence it can be inferred that although for initial cesium intake one can observe an overall weakening of the glass structure, the Q^1 units still remain the major building blocks. For doping beyond 18 mol% cesium definite signatures of disproportionation of the Q^1 units into extended Q^2 and isolated Q^0 units are evident pointing to the collapse of the initial network dominated by the pyrophosphate linkages.

4.3. High pressure studies on IPG

The set of high pressure Raman measurements on undoped IPG from ambient up to 41 GPa along with the decompressed spectrum is shown in Fig. 5a. In keeping with our earlier work where the samples were exposed to a peak pressure of 24 GPa [16] we find that the modes are initially resilient to pressure, but, beyond a certain threshold we observe significant positive, pressure induced shifts. The most intense band at 1082 cm^{-1} attributed to the P–O stretching of the pyrophosphate linkages shifts to about 1120 cm^{-1} at 20 GPa. We attribute the observed positive shifts to a compression of bonds leading to densification of the glassy matrix in response to the applied pressure. The Raman band at 760 cm^{-1}, which is indicative of symmetric stretch of the bridging P–O–P of the pyrophosphate group, is sensitive to changes in both the P–O–P bond length and the inter-tetrahedral bridging P–O–P bond angle [16]. The marginal narrowing of this mode with pressure is associated with a decrease in the spread of the inter tetrahedral bond angle, whilst the hardening of this feature points to decreased inter tetrahedral bond distances leading to compaction of the glass network. The positive shifts become less evident at pressures beyond 6 GPa when the glass matrix becomes highly dense and less compressible. Any further increase in pressure only serves to broaden the Raman modes with a significant loss of intensity that would suggest increased disorder in the matrix. Increased pressure would cause an extensive distribution of bond lengths and bond angles of the phosphate units comprising the glassy network. The overlap of the various closely spaced phonon frequencies would contribute to the width of the modes. Hence the Raman modes would no longer be discrete and would acquire a width due to this pressure induced disorder. No evidence of emergence of new bands of resonance are noted throughout the pressure sweep testifying that no new phases are formed under pressure. The total collapse of the Raman intensities beyond 20 GPa signifies severe distortion of the PO_4 tetrahedra at very high pressures leading to a lowering of the local site symmetry. This could further lead to a reduction of the magnitude of the electronic polarisability of the related vibrations and hence results in an overall loss of Raman intensity [27]. Such a profound loss of intensity has earlier been taken as an evidence of a change in the cation coordination possibly from the tetrahedral to the octahedral state in other tetrahedral glasses [27,28]. Although there exists literature reports that discuss a coordination change for phosphorous under pressure for various specific systems [34–36], in the present work however, we are unable to conclusively remark on the change in the coordination state of the cations without additional experimental evidences. It is to be mentioned that in the present work, IPG has been subjected to very high peak pressures of 41 GPa for the first time, and indeed it is very remarkable to note that the phonons are still observed to bounce back to their ambient position on decompression (Fig. 5a) despite the fact that at high pressures the glass network experiences severe structural distortions. This reversibility suggests that the phosphate backbone that makes up these glasses remains intact and the phosphate chain length too is conserved throughout the entire pressure sweep. This is in sharp contrast to the behaviour of other tetrahedral glasses like silica and germania glasses where permanent densifications and structural transformations occur at pressures as low as 8 and 4 GPa respectively [27].

It is known that the elastic response in glasses to high external pressures is decided by the strength of the constituent bonds, the cation coordination state and the nature of the glass network [27,28,37]. Increase in the number of nonbridging oxygens (NBOs) in the matrix is expected to make the network more compact and hence is known to induce irreversible deformations [37,38]. Although the P–O bonds in phosphate glasses are weaker than the Si–O bonds in silica glass and the number of NBOs in IPG is higher than that in silica glasses, one would expect the elastic response of a compact phosphate glass matrix to be poor compared to silica glasses. On the contrary IPG exhibits unusually high elastic response to high external pressure as compared to silica glass. This exceptional behaviour can be rationalized as follows: Irreversible changes in silica glasses under pressure are attributed to a change in the coordination state of silicon from a 4 fold to a 6 fold coordinated state [28,29] involving permanent/irreversible formation of new bonds in the high pressure state. Whilst silicon is known to exhibit higher coordination states with ease, phosphorous on the other hand does not adopt an octahedral coordination below ~46.3 GPa pressure [39]. Hence in the present pressure sweep of IPG up to 41 GPa, phosphorous prefers to retain its tetrahedral coordination and hence no change to the 6 fold coordination state is envisaged. This rules out the possibility of formation of new bonds in the glass matrix that could lead to permanent plastic deformations. Further, the absence of monovalent ions that would depolymerise the network and the right combination of di- and trivalent cations in IPG make this glass exhibit superior mechanical strength [40]. Hence we conclude that IPG responds to very high external pressures by mere compression of the bonds leading to densifications/tetrahedral distortions that are reversed on decompression.

4.4. High pressure behaviour of IPGC18

Evolution of the Raman spectra with pressure for IPGC18 is shown in Fig. 5b. The sample has been subjected to pressures up to 39 GPa after which it was quenched to ambient. The response of Raman phonons to pressure was similar to that observed in IPG except that the main mode at 1044 cm^{-1} hardens to 1110 cm^{-1} nearly twice the extent as compared to IPG. This greater compressibility of IPGC18 for the same pressure range could be attributed to a weaker network as compared to IPG. Once again at very high pressures (>20 GPa) the Raman intensity is observed to totally collapse pointing to severe distortions and disorder in the matrix. Decompression from the highest peak pressure results in a virtual reversion of all the ambient Raman bands. Whilst in the pressure quenched IPGC18 all the modes revert back to the same position as that of the ambient, small changes are seen with respect to the intensities. In particular the intensity of the 760 cm^{-1} feature attributed to the symmetric P–O–P stretch that bridges the PO_4 tetrahedra shows a small but definite increase in the pressure quenched sample along with a mild hardening (See Fig. 5b). This could mean that pressure not only serves to compress the bonds in the network resulting in the compaction of the PO_4 tetrahedra as is observed in IPG, but also aids in permanent cross linking of the tetrahedral units.

4.5. High pressure studies on heavily doped IPG

Fig. 6 shows the Raman spectral evolution with pressure of the heavily doped glasses — IPGC26 and IPGC35. The high pressure Raman data for IPGC28 is similar to the data of IPGC26 and is hence not shown in the figure. We observe positive pressure induced shifts of the Raman phonons but the extent of hardening is significantly lower as compared to either IPG or IPGC18. With increase in pressure the modes become broad with a systematic fall in intensity, ultimately leading to a featureless spectrum above 20 GPa. The earlier mentioned extra feature at 1184 cm^{-1} due to the P–O symmetric stretch of the Q^1 chain terminator is observed to significantly decrease in intensity with increase in pressure and is seen to completely collapse for pressures beyond 13 GPa. This effect is clearly observed in the highest cesium doped IPGC35 (Fig. 6b). It is known that this feature arises due to a scission of the P–O–P links caused by the cesium dopant leading to depolymerisation of the network. Scission of the P–O–P linkages could lead to the formation of highly reactive charged terminal P–O– species that could be most vulnerable to attack either from the incoming cesium or by its immediate neighbours. Pressure could aid in bringing these reactive species closer, resulting in the relinking of the glass network, ultimately leading to the collapse of the 1184 cm^{-1} mode. It is interesting to note that for pressures greater than 6 GPa a small but definite increase in intensity of the high frequency P–O stretch mode of the Q^2 meta groups in the glass network is observed (see Fig. 6a and b). It now becomes interesting to see whether this increase in Q^2 units could once again suggest a disproportionation of the glass network with pressure (refer Fig. 4). With this in mind, we have carried out a detailed analysis of the high frequency stretch modes of Q^0, Q^1 and Q^2 units in IPGC35 as a function of increasing pressure to look for possible signatures of disproportionation. It is to be mentioned here that for all other samples except IPGC35, the Q^2 modes are lost in the intense diamond feature with increased pressure and hence cannot be followed. The high frequency region of IPGC35 between 850–1260 cm^{-1} has been area normalized and fitted to a mixture of Lorentzian and Gaussian line shapes. The area under each of the fitted components has been extracted and plotted as a function of pressure (Fig. 7). The depletion of the Q^0 units and the increase in Q^1 units in the initial stages of pressurization provide definitive evidence of the isolated Q^0 units dimerising to form Q^1 units. For pressures greater than 6 GPa the dimeric Q^1 units are now seen to give way to the correlated formation of extended chains of Q^2 units. The above analysis thus rules out disproportionation of the network on pressurizing, whilst providing excellent proof of the progressive linking of the phosphate tetrahedra with pressure.

As mentioned earlier, the high frequency Q^1 stretch mode in these heavily doped samples shifts to a much lesser extent than IPG/IPGC18. The trends in frequency shifts of this mode with pressure are clearly brought about in Fig. 8, which shows the variation of mode Gruneisen parameter with cesium content, calculated using the expression $\gamma_{Q^1} = (B/\omega_0)(d\omega/dP)$ in the range of 0 to 20 GPa. Modes are unresolvable beyond 20 GPa. Bulk modulus for IPG is taken as B = 47 GPa [41] and assumed to be weakly dependent on Cs content. From the plot one can identify two distinct classes of compressibility. Samples IPG and IPGC18 exhibit greater compressibility as compared to their heavily doped counterparts — IPGC26, IPGC28 and IPGC35. The inherently higher density of these heavily doped glasses [22], combined with an increase in the extended Q^2 chains in preference to the Q^1 type dimers that evolve with pressure could explain the lower compressibility of these glasses. It is important to note at this juncture that these compact

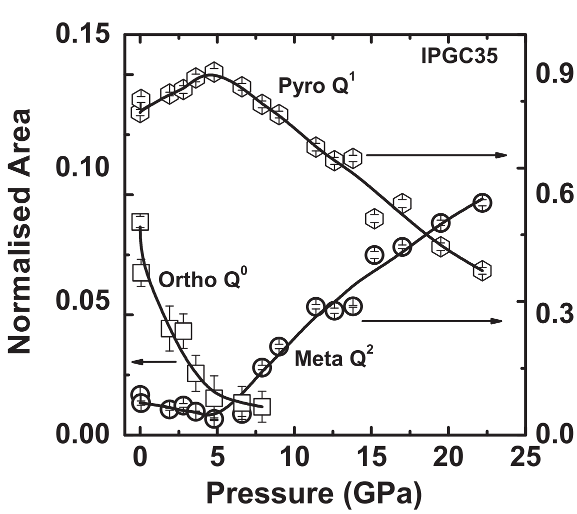

Fig. 7. Variation of normalized area of Q^0, Q^1 and Q^2 units in IPGC35 with increase in pressure. Lines are guide to the eye. Note that in the initial stages of pressurizing a significant drop in Q^0 is observed whilst the Q^1 units are seen to pick up. Beyond 6 GPa the Q^1 units are seen to decrease with a correlated increase in Q^2 units. This is excellent evidence of a progressive linking of the phosphate tetrahedra.

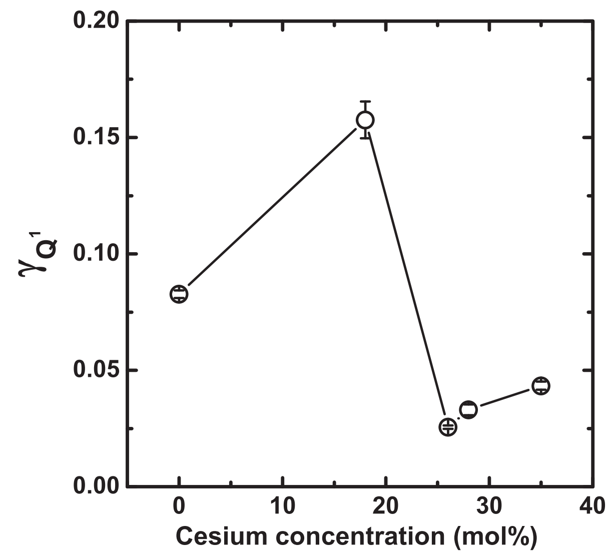

Fig. 8. Variation of mode Gruneisen parameter with cesium content for the dominant P–O stretching mode of Q^1 units calculated using the expression $\gamma_{Q^1} = (B/\omega_0)(d\omega/dP)$ in the range of 0 to 20 GPa; bulk modulus B = 47 GPa [41]. Line is guide to the eye. Heavily doped samples form a less compressible group compared to IPG and IPGC18 that respond significantly to pressure in reversible manner.

heavily loaded glasses that exhibit lower compressibility are the ones that undergo irreversible changes on pressure quenching (Fig. 6a, b). Decompression from the highest peak pressure results in a significantly altered spectrum, both with respect to intensity and peak position in each case — IPGC26, IPGC28 and IPGC35.

In order to look at the changes associated with pressure quenching the sample in greater detail we have plotted the area normalized

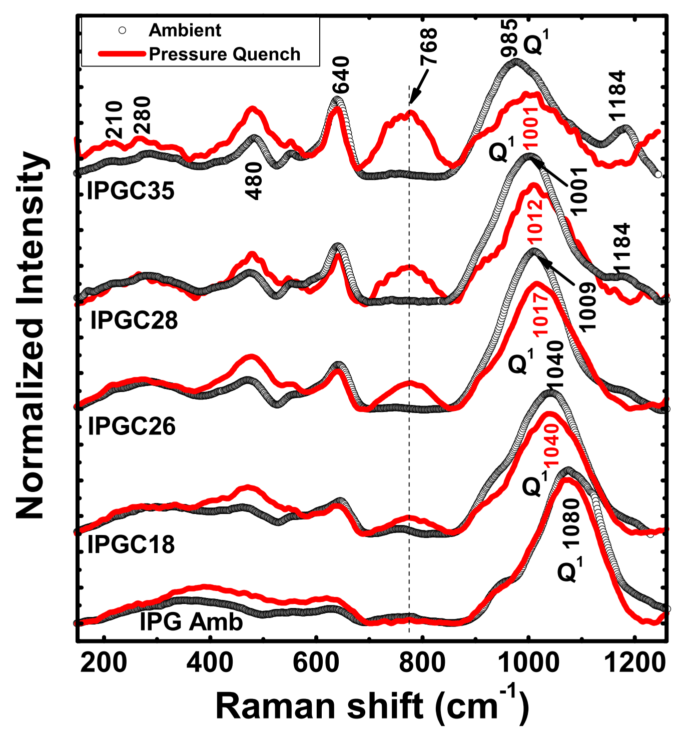

Fig. 9. Set of area normalized ambient spectra along with its pressure quenched counterpart for all the samples. Grey circles represent ambient spectra and red lines represent pressure quenched spectra. Note a systematic increase in the area under the 760 cm^{-1} mode in the pressure quenched samples with increased cesium loading. The positions of the Q^1 mode are marked in each case to clearly denote the shift in the quenched spectra as compared to respective ambient counterparts.

spectrum of the sample under ambient conditions along with its pressure quenched counterpart for each composition (Fig. 9). The extent of irreversibility of the quenched spectra scales well with cesium content, being the maximum for IPGC35. This is once again in contrast to that observed in IPG/IPGC18 where the decompressed spectrum is virtually similar to that of the uncompressed glass. The intensities of the most intense spectral feature corresponding to the stretch of the Q^1 units in these heavily doped glasses in all the decompressed spectra are appreciably decreased as compared to their respective ambient spectra (Fig. 9) and this decrease is attributed to severe tetrahedral distortions of the PO_4 tetrahedra under very high pressure leading to total disorder and irreversible loss of local site symmetry on decompression. Further this intense mode in each of the decompressed spectra of IPGC26, IPGC28 and IPGC35 does not revert back to their initial positions, instead they are observed to get quenched at higher energies as compared to their respective uncompressed samples (Fig. 9). This would mean that under the condition of high pressures, reorganization of the glass network towards a denser tetrahedral network is favoured. The extent of densification in the above samples once again scales well with increased cesium contents as seen from the shift of the Q^1 stretching mode in the pressure quenched sample from its ambient position (Fig. 9). A systematic and dramatic increase in the spectral weight of the bridging P–O–P mode at 768 cm^{-1} feature is observed in all the decompressed spectra that excellently correlates with the cesium content (Fig. 9). This could imply increased cross linking in the decompressed sample as compared to the uncompressed sample. The systematic increase in the cross linking with increased cesium uptake (Fig. 9) in the pressure quenched sample proves beyond doubt that cesium in the sample has served to permanently cross link the PO_4 tetrahedra under conditions of high pressure. The increased attractive interaction between the negatively charged non bridging oxygens created by the incoming cesium and the positively charged cesium ions [8] under high pressure conditions could lead to this irreversible bond formation.

In order to gain further insight into the nature of this irreversibility, we have tried to compare the trends in variation of the area of the bridging P–O–P mode at 760 cm^{-1} for all the samples both on doping and on pressure quenching (Fig. 10). The lower panel of Fig. 10 depicts the variation of normalized area of 760 cm^{-1} bridging P–O–P with cesium doping under ambient conditions and upper panel shows its variation in the pressure quenched sample. It is very clear from the figure that pressure induced irreversible changes are associated with the disproportionation of the pyrophosphate network whilst the elastic responses to pressure are confined to compositions that do not disproportionate. It is to be mentioned here that in the case of alkali silicate glasses that are extensively depolymerised, the excess NBOs in the network bring about pressure induced cationic coordination change, leading to irreversible denser form of the glass upon pressure quenching [42]. Going by the analogy, in the heavily doped IPG, that are extensively depolymerised, irreversible changes to the denser forms of the glass could possibly occur due to a change in the coordination of the network cations to a higher state at elevated pressures. Hence, we conclude that the positive shift of the dominant P–O stretch of the Q^1 units and an appreciable increase in spectral weight for the bridging P–O–P stretch mode at 768 cm^{-1} in the decompressed sample, are coherent with an irreversible transformation from a low density amorphous form of the cesium loaded glass to its high density form, possibly associated with a cationic coordination change.

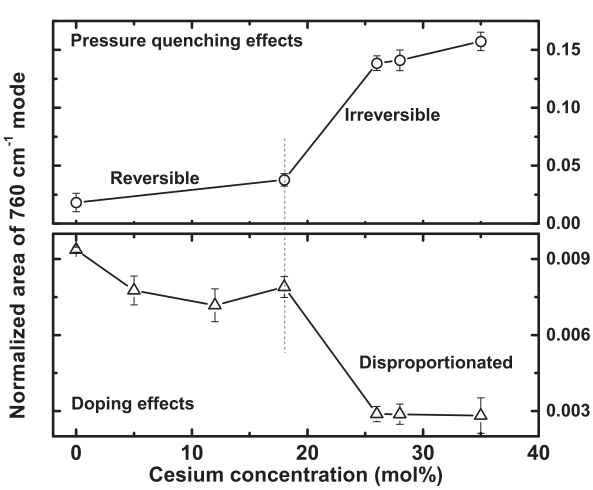

Fig. 10. On the correlation between irreversibility and disproportionation in various Cs doped IPG. Lower panel — Variation of area of bridging P–O–P mode (760 cm^{-1}) with cesium concentration. Upper panel — Variation of area of 760 cm^{-1} mode in pressure quenched sample for each composition. Lines are guide to the eye. Note that pressure induced irreversible cross linking is clearly associated with disproportionation of the parent glass network. Extensive permanent cross linking under pressure is observed in samples that are totally depleted of cross links under ambient conditions.

5. Conclusions

Raman spectra of 40:60 iron phosphate glasses reveal systematic changes in the structural linkages as cesium gets progressively loaded in the matrix. The glass matrix shifts towards a more depolymerised structure with a lower average value of Q^i with cesium addition. Definite signatures of disproportionation of the Q^1 dominated pyrophosphate network into extended Q^2 and isolated Q^0 units are obtained beyond 18 mol% cesium loading. The effect of very high external pressure ~40 GPa on the structural stability of IPG and its cesium loaded analogues has been studied using Raman spectroscopy for the first time. Although signatures of severe structural deformations of the glass network under very high pressure have been obtained, the undoped glass is found to deform elastically and reversibly on decompression to the ambient. This excellent resilient behaviour of the Raman phonons to such high external pressure clearly brings out the superior structural stability of these glasses compared to other tetrahedral framework glasses. The dominant compression mechanism in these glasses could be a gradual reduction in the average P–O–P inter-tetrahedral bond angle and its distribution at low pressures and a compaction/distortion of the PO_4 tetrahedra itself at very high pressures. The tendency of the phosphorous cation to retain its tetrahedral coordination even at very high pressures could have inhibited formation of new bonds aiding reversibility of the transition on decompression. IPG matrix loaded with cesium up to a critical concentration of 18 mol% is also observed to elastically respond to very high external pressure without any major structural modifications. On the other hand, exposing the heavily loaded (>18 mol% cesium concentration) samples to high pressures is found to destabilize the parent network and irreversibly transform the glass matrix into denser forms. Disproportionation of the glass network for cesium doping beyond 18 mol% resulting in the loss of the stable pyrophosphate units, and extensive formation of non bridging oxygens resulting in increased interactions within the network could be the underlying reason that has facilitated the irreversible changes. This reorganization of the parent glass towards denser tetrahedral networks is observed to excellently scale with the cesium content in the sample, thus providing additional proof of the role of cesium in aiding the irreversible changes. The above study clearly demonstrates for the first time that iron phosphate glasses are indeed structurally very stable matrices and can withstand very high pressures without any major structural modifications as long as the pyrophosphate units remain the major building blocks. Samples loaded heavily with cesium are prone to disproportionation and hence are observed to irreversibly transform to denser forms of glass, possibly associated with coordination change of the network cations to a higher state on exposure to elevated pressures.

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